

① a (V-V-F-V)

②

a $0x5C2$
 $\swarrow \quad | \quad \searrow$
 $0101 \quad 1100 \quad 0010$

$000 \ 0101 \ 1100 \ 0010_2$

b 2048×10^6

$$\frac{2 \ 048 \ 000 \ 000}{1 \ 073 \ 741 \ 824} = 1,957 \ 918$$

c

$$\begin{array}{r} 10140100 \\ - 01001101 \\ \hline 01100111 \end{array}$$

③

a -57_{10}

$12/8 \ 64/32 \ 16/8 \ 4/2 \ 1$

00111001

C1: 11000110_2

$$\begin{array}{r} 57 \\ - 32 \\ \hline 25 \\ - 16 \\ \hline 9 \\ - 8 \\ \hline 1 \end{array}$$

d 10011010_2

$\hookrightarrow$ inverted 01100101

$$\left. \begin{array}{l} 0 \cdot 2^7 = 0 \\ 0 \cdot 2^6 = 0 \\ 1 \cdot 2^5 = 32 \\ 0 \cdot 2^4 = 0 \\ 1 \cdot 2^3 = 8 \\ 0 \cdot 2^2 = 0 \\ 0 \cdot 2^1 = 0 \\ 1 \cdot 2^0 = 1 \end{array} \right\}$$

$64 + 32 + 4 + 1 = 101$

$2: -101_{10}$

c) $'0'1'0'10000$

$$+ \begin{array}{r} 01110000 \\ \hline 11000000 \end{array} \quad \text{--- overflow}$$

4)

a) $CPI_{global} = \frac{10^6 \cdot (400 \cdot 1 + 100 \cdot 3 + 150 \cdot 4 + 50 \cdot 4)}{10^6 \cdot (400 + 100 + 150 + 50)} = \frac{1500 \cdot 10^6}{700 \cdot 10^6} \approx 2,142$

b) $T_{cpu} = \frac{CPI \cdot I}{F} = \frac{1500 \cdot 10^6}{2,5 \cdot 10^9} = \frac{1500 \cdot 10^6}{2500 \cdot 10^6} = 0,6x$

c) $CPI_{new} = \frac{10^6 \cdot (400 \cdot 1 + 100 \cdot 3 + 75 \cdot 4 + 50 \cdot 4)}{10^6 \cdot (400 + 100 + 75 + 50)} = \frac{1200}{625} = 1,92$

$$T_{cpu} = \frac{10^6 \cdot (400 \cdot 1 + 100 \cdot 3 + 75 \cdot 4 + 50 \cdot 4)}{2,5 \cdot 10^9} = \frac{1200}{2500} = 0,48x$$

$$\text{speed up} = \frac{0,6}{0,48} = 1,25x$$

5)